To find the problem statement in Assignment-Classification Algorithms

I. A requirement from the Hospital, They want to predict the Chronic Kidney Disease (CKD) based on several parameters. The Client has provided the dataset of the same.

1. LOGISTIC REGRESSION:

- **BEST MODEL WHILE HYPERTUNING LOGISTIC REGRESSION PROBLEM**

PENALTY	SOLVER	MAX_ITER	TOL
L2	liblinear	1000	0.0001

- **CROSS VALIDATION BEST SCORE: 0.9764**
- **ROC_AUC_SCORE: 0.9963**
- **ACCURACY: 0.96**
- **FINAL CLASSIFICATION REPORT:**

	precision	recall	f1-score	support
no	0.95	0.95	0.95	43
yes	0.96	0.96	0.96	57
accuracy			0.96	100
macro avg	0.96	0.96	0.96	100
weighted avg	0.96	0.96	0.96	100

2. SUPPORT VECTOR MACHINE:

- **BEST MODEL WHILE HYPERTUNING SVM CLASSIFICATION PROBLEM**

KERNEL	C	GAMMA
rbf	1	scale

- **CROSS VALIDATION BEST SCORE: 0.9900**
- **ROC_AUC_SCORE: 0.9983**
- **ACCURACY: 0.97**

- **FINAL CLASSIFICATION REPORT:**

	precision	recall	f1-score	support
no	0.95	0.98	0.97	43
yes	0.98	0.96	0.97	57
accuracy			0.97	100
macro avg	0.97	0.97	0.97	100
weighted avg	0.97	0.97	0.97	100

3. DECISION TREE:

- **BEST MODEL WHILE HYPERTUNING DECISION TREE PROBLEM**

CRITERION	MAX_DEPTH	MAX_FEATURES	MIN_LEAF	MIN_SPLIT	SPLITTER
entropy	7	None	1	4	best

- **CROSS VALIDATION BEST SCORE: 0.9766**
- **ROC_AUC_SCORE: 0.9659**
- **ACCURACY: 0.94**
- **FINAL CLASSIFICATION REPORT:**

	precision	recall	f1-score	support
no	0.91	0.95	0.93	43
yes	0.96	0.93	0.95	57
accuracy			0.94	100
macro avg	0.94	0.94	0.94	100
weighted avg	0.94	0.94	0.94	100

4. RANDOM FOREST:

- **BEST MODEL WHILE HYPERTUNING DECISION TREE PROBLEM**

CRITERION	MAX_DEPTH	MAX_FEATURES	MIN_LEAF	MIN_SPLIT	N_ESTIMATORS
entropy	3	log2	1	2	60

- **CROSS VALIDATION BEST SCORE: 0.9898**
- **ROC_AUC_SCORE: 0.9987**
- **ACCURACY: 0.98**
- **FINAL CLASSIFICATION REPORT:**

	precision	recall	f1-score	support
no	1.00	0.95	0.98	43
yes	0.97	1.00	0.98	57
accuracy			0.98	100
macro avg	0.98	0.98	0.98	100
weighted avg	0.98	0.98	0.98	100

Final Model Selection & Justification:

- Both SVC and Random Forest performed exceptionally well, achieving high accuracy, F1-scores and ROC-AUC values. However, **I selected Random Forest as the final model** because it demonstrated slightly better overall performance across the test evaluation metrics.